Multi-View Latent Action Pretraining for Cross-Embodiment Robot Policies

Anonymous Author(s)

Abstract*—Action-free videos are an appealing substitute for costly teleoperated data. Latent action models (LAMs) turn videos into a supervision signal by inferring abstract actions between frames. Yet an action is a 6D motion, and a single camera resolves it only ambiguously missing out-of-plane movements and potential occlusions. We propose a LAM that ingests synchronized multi-camera views and fuses image patches across viewpoints before compressing them into a latent action, raising the decodability of ground-truth 7D actions from *XX* to *XX* R^2. The latent actions feed a policy in which a latent expert and an action expert are coupled by sequential flow matching around a frozen VLM backbone. We evaluate our approach in *cross-embodiment* settings, where task-specific demonstrations come from other robot arms in LIBERO, LIBERO-Plus, and MimicGen, and human-only demonstrations on hardware. Across three simulation suites and *XX* real-robot rollouts, multi-view pretraining improves success rate by *XX%* over single-view LAMs and *XX%* over action-only policies, and we demonstrate transfer from human demonstrations to robot execution under three novel task settings. Beyond these gains, we characterize recurring failure modes that persist when learning from video. *+ highlight that we provide oracle latents to show latent action reconstrution is the bottleneck and not policy performance? Mention continuous latents?

***Index Terms*—Robot Learning, Videos, Scaling, Cross-Embodiment.**

I. Introduction

Data-driven robot foundation models have been shown to scale well with data [1, 2] limiting factor is the availability of robot data to scale, which is often costly to obtain. Recent works propose learning from action-free data, such as videos. Videos contain semantically rich information about tasks that can aid robots in long-term planning.

A challenge is integrating this video information into robot policy learning. A common approach infers a latent action from video with an inverse dynamics model (IDM), and conditions a downstream policy on it [3–5]. The latents actions are an abstract representation describing the dynamics of transitions the video. A key predictor for downstream robot success is how well the latent actions learned from video can be correlated with a real robot action dataset. A number of approaches have been proposed to improve this transferability, including grounding latent actions in real robot actions or states during training[6, 7], or visual pre-processing of the video [8–11]. However, existing single-view approaches cannot recover the full 6D poses required for robot control. In contrast, we propose to leverage multi-view video for latent action pretraining. An action manifests as a 3D motion, which a single view captures only ambiguously — out-of-plane motion is underdetermined, and parts of the scene may be occluded. We therefore propose a Latent Action Model (LAM) that ingests synchronized multi-camera views and fuses image patches across viewpoints before compressing them into a latent action. We demonstrate that spatial awareness leads to better recoverability of 6D robot actions compared to single-view approaches across simulation benchmarks and real-world experiments. Crucially, multiple views are only required during LAM pretraining to improve latent action quality, but not during the policy learning stage where we rely on a single external camera and wrist view.

A key advantage of learning from video is the ability to learn across embodiments, with the ultimate goal of learning from unstructured human videos. Most work uses large datasets for pretraining that confound different embodiments, tasks, and visual diversity [5, 6, 11]. Such uncontrolled settings make it difficult to identify which dataset properties drive policy improvement and under which conditions transfer to new tasks works. Other works only use videos from the target robot embodiment and do not leverage cross-embodiment videos or human demonstrations. We specifically investigate the *cross-embodiment* settings, where task-specific demonstrations are only available from different embodiments. On hardware, the demonstrations come only from a human hand, and in simulation from different robot embodiments. We then add a small amount of paired video and action data from the target embodiment and evaluate performance as we scale the number of target demonstrations. Under this setting, on hardware, we demonstrate the transfer from human demonstrations to robot execution for tasks involving novel movements, scenes, and object placement. On average adding human videos improves task success rate by *XX%* compared to robot-only policies. We find that *add key results for cross-embodiment learning and how this finding relates to other works*.

We propose to combine our LAM pretraining with a lightweight policy-training stage. The policy trains jointly on the small action-labeled dataset and the much larger video-only set using sequential flow-matching expert. The sequential design provides a strong inductive prior, enabling the action expert to effectively leverage the latent plan. The latent-expert predicts the next segment of latent actions, and the action-expert turns that plan into an executable action chunk. A frozen VLM backbone is shared by the two expert providing high-level contextual understanding.

We benchmark our method on simulation suites LIBERO [12], LIBERO-Plus [13], and MimicGen [14] and a real robot platform using *X robot rollouts*. We show that incorporating multiple views substantially enhances the spatial richness of latent actions and downstream

Author(s) Emails.
Sponsorships.

policy performance, and that our method transfers human demonstrations to robot execution on novel tasks involving unseen objects and movements. Beyond these gains, we highlight recurring failure modes in video-based learning and challenges in generalization.

In summary, our main contributions are:

- We propose an approach for IDM-style video pretraining and a downstream policy improving video-free learning on average by $XX\%$ in simulation and $XX\%$ on hardware under cross-embodiment settings.
- We propose to ground latent actions in multi-view videos, leading to XX% better action decodability, and XX% higher policy success rate in sim and XX% on hardware.
- We carefully design experiments on hardware, and demonstrate transfer from human demonstrations to robot execution for novel tasks, and highlight recurring failure cases.

II. Related Work

Learning from Video. Inverse-dynamics models (IDMs) for latent action learning are a major stream for learning from video starting with LAPO [3] and Genie [4]. The LAPO design of coupling an IDM and FDM (forward dynamics model) forms the template of modern LAM approaches. LAPA [5] adopted this approach for robotics by finetuning a VLM on latent actions, and subsequently adding a small MLP head to predict robot actions. villa-X [6] adds robots' proprioceptive states during DIM for physical grounding of the latent actions, while CLAM [7] and LAOM [9] use robot actions for that purpose. However, methods that ground pretraining in robot data might degrade for cross-embodiment settings and are not suitable for learning from human videos. Other works propose continuous latents which appear to work better than discrete tokens for continuous robot control [8–10, 15]. AMPLIFY [16] introduces motion keypoint trajectories and LAOF [10] optical flow as an intermediate representation to learn latent actions from videos. UniVLA [11] relies on DINO features to predict latent actions instead of pixels. Those methods still rely on a single viewpoint and video processing, while we propose to use multiple viewpoints for better action recoverability.

Multi-View Learning. Many robot foundation models operate on a single egocentric viewpoint [2, 17, 18], although recent work shows that 3D spatial perception yields stronger manipulation policies[19–21]. One family of methods injects explicit geometric reconstructions into the policy from point clouds [19], depth maps [22], or voxels. For instance, DP3 [19] encodes reconstructed point clouds with a PointNet [23], and DepthVLA [22] fuses depth through cross-attention. These approaches, however, rely on auxiliary depth sensors or complex reconstruction pipelines that introduce noise. A complementary family avoids explicit 3D inputs and learns geometric structure implicitly leveraging multi-view supervision to pretrain the visual backbone of foundation models: RoboUniView [22] pretrains a unified view representation from multi-perspective images, 3D-MVP [21] pretrains the visual encoder via masked autoencoding over multi-view RGB-D renderings, CLAMP [24] learns visual representations through a multi-view contrastive objective, and [25] trains backbones on cross-view spatial relations that transfer to improved manipulation success. MV-MWM [26] and ReViWo [27] train multi-view encoder-decoders to build viewpoint-robust world models for downstream robot policy learning.

As latent action models aim to encode embodiment-agnostic, higher-level plans, they stand to benefit substantially from 3D-aware pretraining. However, 3D pretraining for latent action models remains largely under-explored. Alongside concurrent work [28], we propose to leverage multi-view pretraining of LAPA-style models to improve latent action learning. Crucially, the multi-view videos are required only during the pretraining and can be drawn from multi-view human videos; downstream policy learning and deployment can proceed from a single view.

Cross-Embodiment Learning. Videos are a natural source for cross-embodiment learning from human demonstrations. One approach pools demonstrations from different robots and absorbs their heterogeneity architecturally through per-embodiment input tokenizers [18, 29], separate action heads [2, 30], or embodiment identity tokens [31] around a shared trunk. Other works leave the architecture untouched and instead close the *visual* gap by editing pixels through inpainting at test time [32], inpainting at train time [33], or segmentation overlays [34]. The same method can bridge the human–robot gap by rendering a robot at the estimated hand pose [35–37]. These methods make the mechanism of transfer interpretable, but depend on URDFs, renderers, or hand-pose estimators.

Closest to us, XIRL [38] and XSkill [39] discover embodiment-agnostic skills from unlabeled human and robot video. UniAct [40] decodes a vector-quantized codebook of universal behaviors, and [41] contrastively aligns end-effector actions across hands and grippers. Latent action models reach a similar interface from video alone, without paired or retargeted data [5, 11, 42]. Our LAM takes the weakest supervision assumption requiering neither action labels, proprioception, or a retargeting function.

The transfer question remains confounded. Work that leverages aggregated datasets such as OXE [43] vary embodiment, task, scene, and visual diversity all at once [5, 6, 44]. Those settings make it difficult to verify if task skills are transferred between morphologies. We adopt a controlled protocol in which task-specific video demonstrations stem from *different* robot morphologies in LIBERO [12], LIBERO-Plus [13], and MimicGen [14], and exclusively from human demonstrations on hardware. We then scale the fraction of data we add from the target embodiment, and investigate which task settings transfer.

III. Method

We consider vision-based manipulation under heterogeneous action supervision: a small set of action-labeled demonstrations on a target embodiment and a much larger set of cross-embodiment video-only demonstrations. Our method has two stages. A latent action model (LAM) is pretrained on videos to infer a compact *latent action* for each observed transition (Sec. III-B), using multiple viewpoints to make those latents action-centric (Sec. III-C). A policy is then trained by imitation on both sources jointly: it predicts a latent *plan* from context and an executable action chunk conditioned on that plan (Sec. III-D). The latents serve purely as policy-internal planning variables, and the deployed policy interface is the robot actions.

A. Problem Formulation

We consider language-conditioned visual manipulation. Given a visual observation $o_t \in \mathcal{O}$ and a task instruction ℓ, the agent acts in the robot's action space $\mathcal{A}$. We seek a policy π_θ that predicts an action chunk $a_{t:t+K}$ of K future actions $a_t \in \mathcal{A}$ from the current context $x_t = (o_t, \ell)$.

We learn this policy by imitation. The standard imitation objective for learning from a dataset $\mathcal{D}_a = \{(\ell^{(j)},\, o^{(j)}_{0:T_j},\, a^{(j)}_{0:T_j})\}_j$ of action-labeled trajectories minimizes the behavior cloning loss

$$\min_\theta\ \mathbb{E}_{(x_t,\, a_{t:t+K})\sim\mathcal{D}_a}\big[\mathcal{L}_{\mathrm{BC}}\big(\pi_\theta(x_t),\, a_{t:t+K}\big)\big].$$

In our setting, the action labels are scarce and $\mathcal{D}_a$ holds only a small number of demonstration trajectories. Additionally, we have a second, action-free dataset $\mathcal{D}_v = \{(\ell^{(k)},\, o^{(k)}_{0:T_k})\}_k$ holding a much larger set of video-only demonstrations of cross-embodiment systems. The problem we address is learning a policy π_θ that leverages both datasets, $\mathcal{D}_v$ and $\mathcal{D}_a$.

B. Latent Action Model

Following the inverse/forward-dynamics approach for action-free video [3–5], the LAM couples an inverse dynamics model (IDM) that infers a *latent action* from observed transitions with a forward dynamics model (FDM) that must reconstruct the future from the present and that latent.

Unlike the common formulation ingesting only one current and one future frame of a single viewpoint [3, 5], our IDM ingests a multi-view window $O_t = (o_{t+\delta})_{\delta\in\Delta}$ over frame offsets $\Delta = \{0, \delta_1, \ldots, \delta_m\}$ (a current frame and m future frames) and encodes it jointly with a multi-view spatiotemporal encoder Φ (detailed in Sec. III-C), producing for each frame $\delta \in \Delta$ a fused token map $f_{t+\delta} = \Phi_\delta(O_t)$. A small bottleneck B compresses each map. The latent action for offset δ is the difference of bottleneck codes of the future and current frames, producing $|\Delta| - 1$ latent actions $z_t^{(\delta)}$:

$$z_t^{(\delta)} \;=\; B\big(\Phi_\delta(O_t)\big) \;-\; B\big(\Phi_0(O_t)\big). \tag{1}$$

The FDM reconstructs each future frame $o_{t+\delta}$ from the current-frame tokens $e_t = E(o_t)$ — the patch embedding E shared with the encoder Φ — and the corresponding latent $z_t^{(\delta)}$, trained with a robust reconstruction loss ρ (Huber loss):

$$\mathcal{L}_{\mathrm{LAM}} \;=\; \mathbb{E}_t \sum_{\delta\in\Delta,\,\delta>0} \rho\Big(\,\mathrm{FDM}\left(\mathrm{sg}[e_t],\, z_t^{(\delta)}\right),\; o_{t+\delta}\Big). \tag{2}$$

The stop-gradient $\mathrm{sg}[\cdot]$ on the current-frame tokens forces the only trainable path that lowers future-reconstruction error to run *through the latent* (and not through E), so z captures the controllable change rather than static scene content.

During training, we differently augment the IDM input and the reconstruction target, which further encourages the latent to be invariant to appearance and cannot encode it as a shortcut.

A single joint pass of Φ over the window Δ yields a latent at every horizon at once, each supervised by its own future reconstruction in Eq. (2). We let the policy consume only one of the latent at its planning horizon H while the remaining offsets only regularize the latent action learning.

C. Multi-Viewpoint Fusion

An action manifests as 3-D motion of the end-effector and objects, which a single RGB view captures only ambiguously — from any single viewpoint out-of-plane motion are underdetermined, and parts of the scene may be occluded. We hypothesize that multiple orthogonal and synchronized viewpoints give the LAM a better 3-D understanding of the scene and hence more action-centric latents than a single view, supplying better spatial grounding.

Multi-view spatiotemporal encoder. Each RGB frame $I_t^{(v)}$ (view $v \in \{1, \ldots, V\}$, window offset $t \in \Delta$) is split into N patches; patch p of view v is embedded by the shared patch embedding E and tagged with a learned camera-identity embedding c_v, giving tokens $\mathbf{x}_{t,v,p}$. These pass through a stack of spatial attention blocks followed by a stack of temporal attention blocks. We borrow the factorized space-time attention that has proven effective for video understanding in computer vision [45, 46] and adapt it similarly to recent world models [4, 26], to the multi-view setting. Every block uses ℓ_2-normalized (cosine) multi-head attention [47] with rotary position encodings (RoPE) [48],

$$\begin{aligned}\mathrm{Attn}(Q,K,V) &= \mathrm{softmax}\big(s\,\hat{Q}\hat{K}^\top\big)\,V,\\ \hat{Q} &= \gamma_q \odot \frac{Q}{\|Q\|_2},\\ \hat{K} &= \gamma_k \odot \frac{K}{\|K\|_2},\end{aligned} \tag{3}$$

with a fixed temperature s and learned per-dimension scales γ_q, γ_k; rotary embeddings are applied to $\hat{Q}, \hat{K}$ before the inner product.

Joint cross-view attention. We let the *spatial* block attend over the union of all views' patches — so a patch in one view attends to patches in every other view for a given timestep t — while the *temporal* block attends across frames at each fixed location; views are then pooled into the fused map that defines the latent:

$$\begin{aligned}\text{cross-view spatial:}\quad & \mathbf{x}_{t,\cdot,\cdot} \;\leftarrow\; \mathrm{Attn}\left(\{\mathbf{x}_{t,v,p}\}_{v\in[V],\,p\in[N]}\right),\\ & \text{2-D RoPE on } p;\\ \text{temporal:}\quad & \mathbf{x}_{\cdot,v,p} \;\leftarrow\; \mathrm{Attn}\left(\{\mathbf{x}_{t,v,p}\}_{t\in\Delta}\right),\\ & \text{1-D RoPE on } t;\\ \text{view fusion:}\quad & f_t \;=\; \tfrac{1}{V}\textstyle\sum_{v=1}^{V} \mathbf{x}_{t,v,\cdot}.\end{aligned} \tag{4}$$

Encoding the views *jointly* — rather than with independent per-view encoders and late fusion [49, 50] — lets the latent exploit cross-view correspondence at every layer. This factorization suits our setting: because the viewpoints are synchronized and static, patches across views correspond to shared

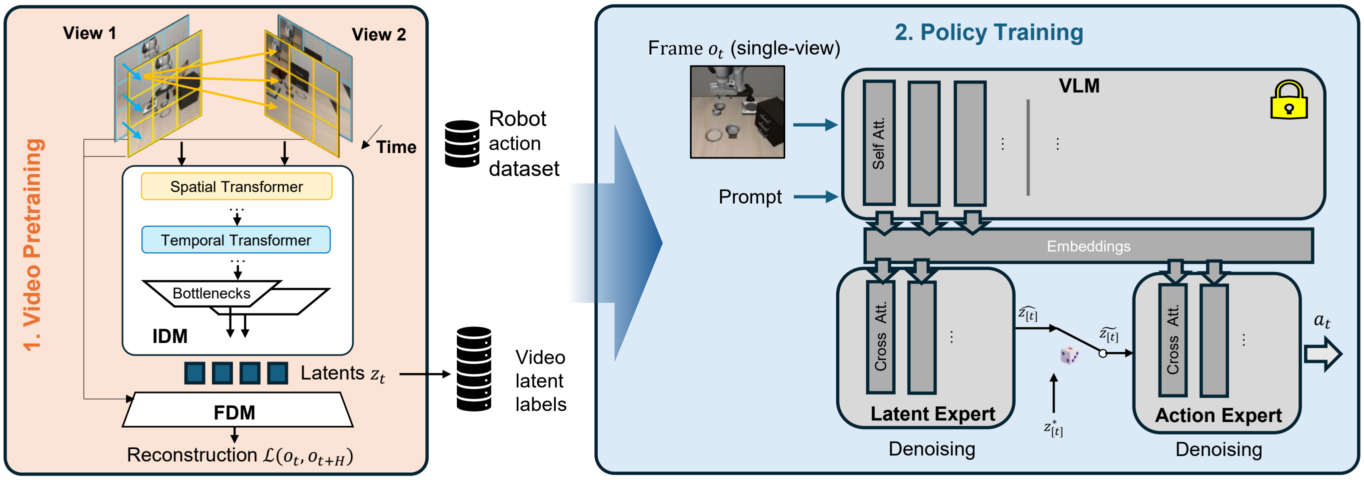

Fig. 1: Method overview.

3-D structure at each instant, so cross-view attention in the *spatial* stage resolves per-frame geometry (depth, occlusion) while the *temporal* stage isolates how that geometry changes via the action. The resulting maps $f_{t+\delta} = \Phi_\delta(O_t)$ feed the difference latent of Eq. (1).

We note that the multiple context viewpoints can provide privileged information to train the LAM and label the video data. The same viewpoints do not necessarily have to be available for downstream policy training.

D. Latent-Augmented Policy

We design a hierarchical policy that first predicts a latent plan from context, then an action chunk conditioned on that plan, factorizing the joint distribution as

$$p_\theta\big(a_{t:t+K},\, z_{[t]} \mid x_t\big) \;=\; \underbrace{p_\theta\big(z_{[t]} \mid x_t\big)}_{\text{latent plan}} \cdot \underbrace{p_\theta\big(a_{t:t+K} \mid x_t, z_{[t]}\big)}_{\text{action chunk}}, \tag{5}$$

where $x_t = (o_t, \ell)$ is the context. Mirroring the action chunk of K actions, the latent plan $z_{[t]} = (z_{t+jH})_{j=0}^{M-1}$ is a chunk of $M = K/H$ latents spanning the same interval, but with fewer entries since each latent covers H frames — the single planning horizon retained from the window Δ (Eq. (1)). This differs from joint latent–action models [6]: the action depends on the plan through an explicit conditioning interface, not a shared denoising target.

We build the policy on a frozen vision-language backbone with lightweight expert transformers in the manner of recent flow-matching VLA policies [2, 51]. Two flow-matching experts attend to the same backbone and run in sequence: a latent expert predicts the plan, and an action expert predicts the chunk conditioned on the plan. Inspired by SmolVLA [51], we structure the backbone–expert interaction as blocks that alternate cross-attention to the backbone's keys and values — projected into the expert's width — with self-attention over the expert's own tokens. We use this interface for *both* experts, so that each one retains the backbone's pretrained knowledge. Each expert is trained by conditional flow matching: for a target y (a latent plan or action chunk) with conditioning c, noise $\varepsilon \sim \mathcal{N}(0, I)$ and time τ,

$$\begin{aligned} x^\tau &= \tau\varepsilon + (1-\tau)y, \quad u = \varepsilon - y, \\ \mathcal{L}_{\mathrm{fm}}(y, c) &= \mathbb{E}_{\tau,\varepsilon}\big\|v_\theta(x^\tau, \tau, c) - u\big\|^2, \end{aligned} \tag{6}$$

with denoised estimate $\hat{y} = x^\tau - \tau\, v_\theta(x^\tau, \tau, c)$. The two experts differ only in their conditioning: $c_z = x_t$ for the latent expert and $c_a = (x_t,\, z_{[t]})$ for the action expert.

Supervision. Latents are inferred from video alone, so every sample of $\mathcal{D}_v$ and $\mathcal{D}_a$ carries a latent target, while only $\mathcal{D}_a$ carries action labels. We formulate a loss that enables action-free video and action-labeled demonstrations to jointly train one policy:

$$\mathcal{L} \;=\; \lambda_a\, \mathbb{E}_{i\sim\mathcal{D}_a}\big[\ell_a^i\big] \;+\; \lambda_z\, \mathbb{E}_{i\sim\mathcal{D}_v\cup\mathcal{D}_a}\big[\ell_z^i\big], \tag{7}$$

where ℓ_a^i, ℓ_z^i are the per-sample flow-matching losses of Eq. (6) averaged over chunk steps and dimensions, and λ_a, λ_z are weights.

However, the action expert faces a *train–deployment mismatch* — at training it could condition on ground-truth plans, but at deployment only on its own prediction. We close this gap using scheduled sampling [52]: the action expert is conditioned on the ground-truth plan $z^\star$ with probability p and otherwise on its own predicted $\hat{z}$, $\tilde{z} = m\, z^\star + (1-m)\, \hat{z}$, $m \sim \mathrm{Bernoulli}(p)$, with p decaying over training. At deployment $\hat{z}$ is available and used.

IV. Training and Implementation Details

Table IX lists the model and training hyperparameters. Both stages train on NVIDIA H100 GPUs: the LAM for 21 h on four GPUs, the policy for 21 h on one. The LAM is trained on all available videos, which mostly consist of cross-embodiment data. The policy is latent-supervised on the same videos and action-supervised for the available trajectories on the target embodiment.

Latent action model. The encoder is trained from scratch, with 8 spatial and 8 temporal attention layers, over $V = 2$ synchronized views. It ingests the frame window $\Delta = \{0, 1, 5, 9\}$ and produces 1×8 latents; the policy consumes only the

H=5 latent, i.e. a 0.25 s horizon at the 20 Hz frame rate. We train it for 70k steps. The appearance augmentation of Sec. III-B includes random crop, brightness, saturation, and contrast jitter.

Policy. We use a frozen SmolVLM2 vision-language backbone and predict a plan of M=10 latents followed by an action chunk of K=50, trained for 100k steps. The policy does not condition on proprioceptive state. At deployment it runs receding-horizon control by executing the first 10 actions of the chunk, and then re-computes.

Latent Action Probing. Latent actions should encode information that can be executed on the real robot. Thus, real robot actions should be reliably *predictable* from the latent actions. Prior works [**...**] show that *probing* - training a lightweight MLP to predict ground-truth robot actions from frozen latent embeddings - provides a reliable metric for evaluating the quality of the learned latent space and predicting downstream policy performance. We report the coefficient of determination,

$$R^2 = 1 - \frac{1}{D}\sum_{d=1}^{D}\frac{\sum_i (a_{i,d}-\hat{a}_{i,d})^2}{\sum_i (a_{i,d}-\bar{a}_d)^2}, \quad \hat{a}_i = f_\phi(z_i), \qquad (8)$$

where f_ϕ is the frozen-input MLP head. Because f_ϕ is intentionally low-capacity, it cannot learn complex behavior on its own; its accuracy therefore reflects how well the *fixed* latent already predicts robot actions (R^2=0 corresponds to predicting the per-dimension mean action).

Oracle Latents. To estimate an upper bound of policy performance given perfectly reconstructed latent actions, we introduce *oracle latents*. The oracle latents are defined directly from the available ground-truth 7D actions a_t

$$z_t^{\text{oracle}} = \sum_{k=0}^{H-1} a_{t+k},$$

and we train the identical downstream policy on it. z_t^{oracle} estimates perfect reconstruction, and thus the added value of the video dataset in the best scenario. It serves as an upper boundary that the video can provide to downstream policy learning. Any remaining gap to expert performance is attributable to how the policy leverages latent actions rather than to LAM reconstructions.

V. Benchmarks & Dataset Splits

For the benchmarks LIBERO [12], LIBERO-Plus [13], and MimicGen [14], we generate demonstrations across five embodiments (Panda, Sawyer, IIWA, UR5e, Kinova3). The panda robot is the target embodiment, while the other cross-embodiment demonstrations only ever contribute video data. The results of the data generation are provided in Table VII. The hardware setting is explained in subsection V-D

A. LIBERO

LIBERO is a benchmark with four task suites: Spatial, Object, Goal, LIBERO-10. Each suite contains 10 tasks, and each task contains up to 50 demonstrations in the dataset.

- **Spatial**: *"Seen object, novel layouts"*. The task goal is the same for all tasks ("pick and place the bowl"), but the scene layout changes.
- **Object**: *"Seen layout, novel objects"*. The goal is to place different objects ("soup", "ketchup" in a basket across 10 tasks.
- **Goal**: *"Same objects and layout, different goals"*. The scene layout is always the same, but the task goals differ, and which objects to handle ("place mug", "open drawer").
- **LIBERO-10**: *"Complex long-horizon tasks"*. Long horizon tasks with diverse task goals, objects, and scene layouts.

We generate the dataset by replaying the source demonstrations on the target embodiments, yielding the results in Table VII. The process yields 6,015 successful episodes across the five embodiments, of which 1,722 are Panda. LIBERO-90 adds a further 13,338 episodes (3,936 Panda), used as play data, bringing the generated corpus to 19,353 successful episodes over 112 tasks.

We scale the number of per-task action-annotated demonstrations for the Panda between 1 and 20. In addition, two task from each LIBERO suite are *holdout tasks* throughout all experiments that never see any robot action data during training to test generalization under video-only scenarios. The task-split is given in Table VIII.

B. LIBERO-plus Suite

LIBERO-plus [13] is an extension of LIBERO [12] that adds perturbation factors: distractor objects, camera viewpoint, lighting, background texture, camera noise, and language paraphrase. The generated data corpus spans 73.2k episodes, of which 15.4k are Panda.

For the action-annotated demonstrations for the Panda, we use 2 demonstrations per task-disturbance combination, i.e., 5% of the total LIBERO-plus Panda action data available. The other other embodiments enter the dataset purely as video demonstrations.

C. MimicGen Suite

MimicGen [14] is an automated demonstration-generation system built on Robosuite/MuJoCo [53]. We use the complete core suite of 28 task variants, generating demonstrations rendered at 256×256. Retaining only task-embodiment combinations with $> 20\%$ generation success rate yields 107 of 140 combinations and 42.3k episodes, of which 12,7k are Panda.

Similarly to LIBERO, we use a large portion of the dataset as play data not used for evaluation, and evaluate on 6 separate eval tasks Figure 2. For the eval tasks, we use 5% of the available demonstration (5 episodes per task) for the target embodiment. Reported are success rates over 50 rollouts per task.

D. Real-World Experiments

In the real world, we test embodiment transfer from human to robot. In total, we collect 1000 human and 600 robot

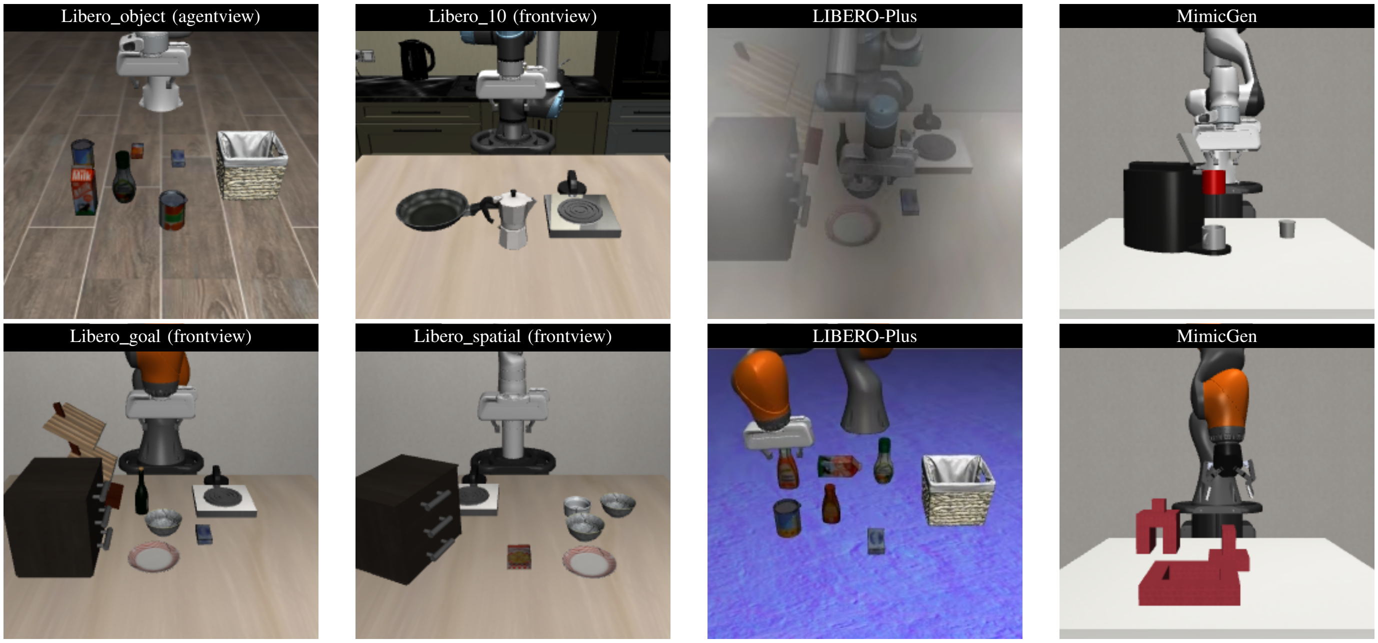

Fig. 2: Overview of the benchmark suites. **LIBERO:** The four LIBERO benchmark suites. Front- and sideview are only used for visual pretraining. The agent- and a wristview are used for policy training. The embodiments include Franka Emika Panda, UR5e, KUKA LBR iiwa, Kinova Gen3. **LIBERO-Plus** introduced visual disturbances to the LIBERO tasks, such as camera noise and different textures. **MimicGen:** For all 28 MimicGen tasks, we generate demonstrations for five embodiments and for front-, side-, agent-, and wristview cameras. Shown are two evaluation tasks: coffee preparation and three piece assembly.

episodes across 32 distinct tasks as play data (Figure 7). We then construct a new task set for evaluation that is not present in the play data. We collect another 50 human and 50 robot demonstrations per task, and train policies with varying number of human and robot data for evaluation.

The experiments use a TRLC-DK1 [54] robot, running in end-effector space control. The policies run in synchronous inference mode at 30 fps, execute the first 10 actions of the chunk, and use video resolutions of 480×640.

All policy evaluations use 25 rollouts for each task, amounting over 550 real-world policy rollouts. The positions of all objects in the scene are randomized for each rollout.

VI. Evaluation

A. Policy Performance

Goal: show that our method improves above baselines across simulation benchmarks. Learning from video experiments use the 5% action labels. baseline: using available action data already during LAM training. Note: can also note down traiing times, since more input frames = more training.

B. Scaling With Data

Here go non-heldout (in-dist.) tasks. It tests "If I have data for the same tasks, how well does video data help". It mainly tests robustness towards objects placements, in our case.

Here go heldout (OOD) tasks. It tests "If I have video data for different tasks, how well can I generalize to entirely new tasks". It tests deeper semantic understanding of the learned latents. Divided into two analysis: depth sweep vs breadth sweep.

TABLE I: Rollout success rate (%) across the three benchmarks. The *LAM* column marks whether the policy receives latent-action supervision (✓) or action supervision only (–).

Method	LAM	MimicGen	LIBERO	LIBERO+
Full action budget	–	65.3	TBD	85.1
Low action budget				
Ours (policy only)	–	25.7	TBD	49.8
$\pi_{0.5}$	–	32.0	TBD	54.0
villa-X (state grounding, VQ latent)†	✓	18.8	TBD	45.1
villa-X (state grounding, cont. latent)†	✓	22.0	TBD	51.5
UniVLA-style (DINOv3 features)‡	✓	28.7	TBD	60.9
CLAM-style (action grounding)‡	✓	32.3	TBD	63.1
Ours (single-view)	✓	**31.0**	**TBD**	**62.1**
Ours (multi-view)	✓	**38.7**	**TBD**	**66.7**
Oracle*	(✓)	39.2	TBD	72.43

†Pretrained latent-action model used frozen. ‡Re-implementation on our architecture based on authors' released code, so that only the named component differs from ours. *Oracle latent actions computed directly from ground truth actions.

Qualitative Analysis. *Show casees for which generalization works and for which it doesnt, and the reasons for it.*

C. Scaling with Cross-Embodiment Data

D. Failure Cases

Show (quantitatively and qualitatively) where learning from video fails.

E. Analysis of Latent Quality

We show that our metric correlates with policy performance, and that multi-view videos are required for accurate reconstruction of 3D motions.

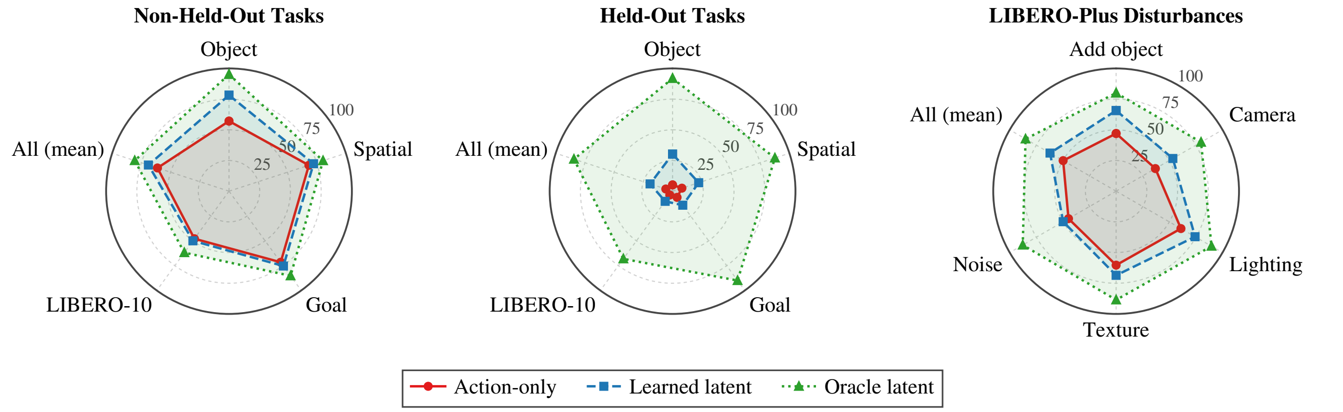

Fig. 3: Per-suite success rate breakdown at the 5% action budget. **Left:** LIBERO tasks that receive action-labelled demonstrations. **Middle:** held-out LIBERO tasks, which never do (depth sweep). **Right:** LIBERO-Plus, broken out by disturbance dimension. Each spoke is one suite or one disturbance; *All (mean)* aggregates the spokes to its left. The oracle latent stays near the rim throughout while the learned latent collapses inward as the setting gets harder.

	# Episodes with action labels			
Policy (SR [%])	**1**	**5**	**10**	**20**
Action-only	43.7	63.8	68.4	74.5
+ LAM (ours)	52.5	69.8	71.7	71.2
Holdout tasks (never seen action data)				
Action-only	2.9	1.9	1.6	5.1
+ LAM (ours)	39.4	43.2	55.1	50.2

TABLE II: Policy performance measured as Success Rate (SR [%]) with increasing video dataset size relative to robot dataset size. Bottom block reports performance on holdout tasks, which never receive action-labeled episodes and are evaluated purely under video-only generalization.

	% of LIBERO-90 used as play data			
Policy (SR [%])	**0**	**33**	**66**	**100**
Action-only	62.2	59.6	?	68.5
+ LAM (ours)	66.6	64.1	?	65.5
Holdout tasks (never seen action data)				
Action-only	0.0	0.5	?	6.9
+ LAM (ours)	27.9	41.8	?	49.3

TABLE III: Policy performance measured as Success Rate (SR [%]) with increasing play-data diversity (fraction of LIBERO-90 used for video pretraining), at a fixed action budget of X episode per task. Bottom block reports performance on holdout tasks, which never receive action-labeled episodes and are evaluated purely under video-only generalization.

F. Relation to Related Results

Could look into how our results relate to performance gain when learning from video for other papers. They often use different architectures and datasets making direct comparison difficult, but we could compare our gains to theirs. Especially

	# X-embodiments in video pretraining		
LIBERO tasks (SR [%])	**1**	**2**	**4**
Non-held-out	63.6	65.7	69.7
Held-out	58.3	49.3	58.9

TABLE IV: Effect of cross-embodiment video diversity on LIBERO policy performance. The column heading counts the *additional* embodiments whose videos enter latent-action pretraining, drawn from {Sawyer, IIWA, UR5e, Kinova3}. All entries are the LAM policy (ours) at X episodes action-budget.

Rollout outcome	**w/ Video (ours)**	**Action-Only**
success	60.5%	37.5%
Failure cases		
placement or collision	14.0%	9.5%
grasp or transport	20.5%	31.0%
coarse approach	2.5%	14.0%
wrong plan	2.5%	8.0%
Total	**100%**	**100%**

TABLE V: Failure case analysis. Add multi-view

interesting: large scale pretraining or not? This also concerns learning on subsets of libero datasets.

G. Real World Experiments

- **Novel movement**: The push task needs high precision. A majority of failure cases comes from pushing the object off-center leading to misalignment of the object. , since pushing the object off-center leads to misalignment of the object. It is also the only non-prehensible manipulation task, which makes it hard to generalize for video only data. All other task perform some kind of grasping. While the LAM extracts general movements, hand actions are difficult to replicate also due to embodiment differences.
- **Novel background**: We were surprised to find significant performance degradation for changing the background. Notably the same task has been in the play dataset with

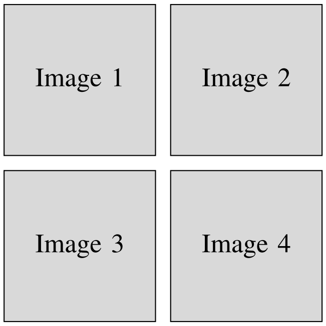

Fig. 4: Image renderings of failure cases for LIBERO to visualize Table V.

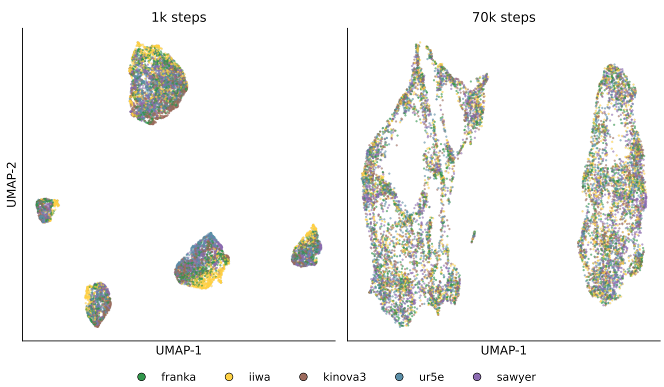

Fig. 5: Real world tasks scaling.

the standard white background. Our given demonstrations are insufficient to compensate to background changes. The model potentially overfits to the white background used in the rest of the data.

	Novel obj		Novel place	
Rollout outcome	**w/ Video (ours)**	**Action-Only**	**w/ Video (ours)**	**Action-Only**
success	34.0%	—	58.0%	—
Failure cases				
missed grip	44.0%	—	14.0%	—
dropped after grip	2.0%	—	0.0%	—
target location error	0.0%	—	2.0%	—
unwanted collision	0.0%	—	0.0%	—
grip far off	20.0%	—	26.0%	—
wrong plan	0.0%	—	0.0%	—
Total	**100%**	—	**100%**	—

TABLE VI: Real-world failure analysis for the 5-robot/50-human-episode setting. Each generalization category pools two tasks (50 rollouts). Pictures are shown in Figure 10.

VII. Discussion

- We observe a large discrepancy between oracle and video latents. This indicates that not policy training but latent action pretraining is the current bottleneck for downstream performance. With multi-camera view, we already suggest one measure to improve video pretraining. Future work should investigate other ways, such as XXX.

VIII. Conclusion

Acknowledgments

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

Appendix A
Data Generation

TABLE VII: Demonstration generation successes per embodiment. LIBERO replays the existing 6.500 source demonstrations, while MimicGen resamples until the success episode quota is met. LIBERO has a total of 112 tasks, MimicGen 28. LIBERO-Plus is an extension of LIBERO, and thus the data generation mechanism is the same. A task counts as covered if the embodiment yields at least one successful demonstration for it. **SR:** success rate, **# eps:** number of episodes generated, **# tasks:** number of different tasks covered.

Embodiment	**LIBERO**	**MimicGen**
	SR [%] / # eps / # tasks	
Panda (target)	87.0 / 5,658 / 111	40.8 / 2,700 / 27
IIWA	72.4 / 4,709 / 101	29.5 / 2,100 / 21
Sawyer	10.5 / 680 / 33	26.1 / 2,000 / 20
UR5e	64.2 / 4,176 / 101	40.4 / 2,400 / 24
Kinova3	63.5 / 4,130 / 102	23.7 / 1,500 / 15
All	59.5 / 19,353 / 112	32.7 / 10,700 / 28

Appendix B
Failure Cases

(a) Grasp far off. (b) Missed grasp.

(c) Wrong placement. (d) Missing precision.

Fig. 10: Reoccuring failure modes for policies.

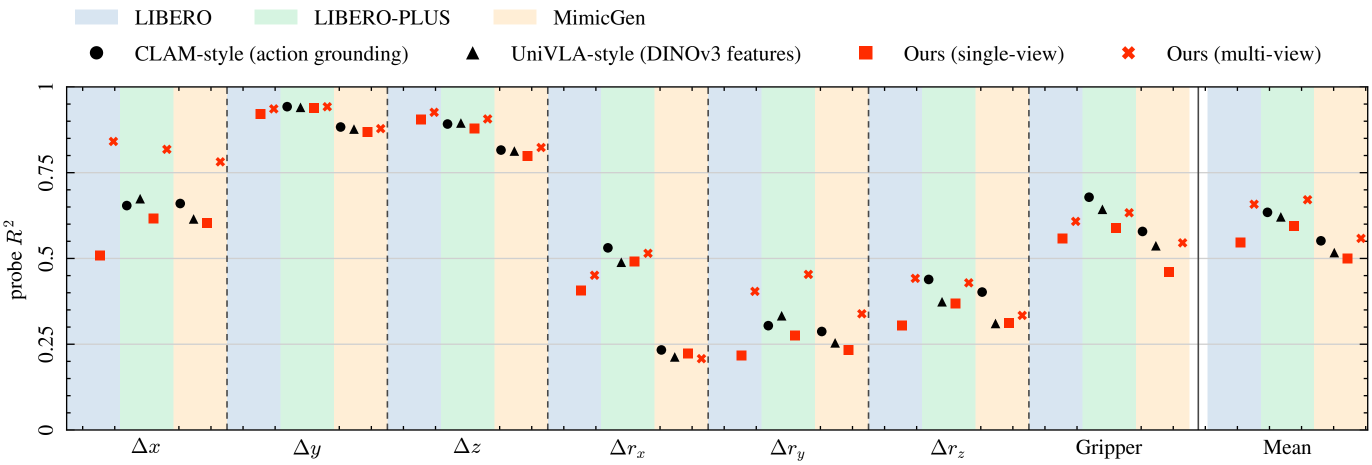

Fig. 6: Latent action quality: A MLP probe reconstructs ground-truth robot actions from learned latent action labels; we report per-dimension R^2 (higher is better). **Mean** aggregates over all seven action dimensions.

Fig. 7: Real world tasks: we collect 1000 diverse human demonstrations and 600 robot demonstration on a TRLC-DK1 [54] robot accross different tasks. A separate task suite consisting of six novel tasks is used for evaluation, where we collect 50 human and 50 robot demonstrations each. On these tasks, we test generalization from human to robot for novel movements, novel background, novel objects, and novel placement locations.

Appendix C
LIBERO Task Suite Mapping and Supervision Splits

TABLE VIII: Task Suites and Corresponding Holdout Task Instructions

Task Suite	Task Names of Holdout Tasks
LIBERO-Goal	`turn on the stove,` `put the bowl on the plate`
LIBERO-10	`put the black bowl in the bottom drawer of the cabinet and close it,` `pick up the book and place it in the back compartment of the caddy`
LIBERO-Spatial	`pick up the black bowl from table center and place it on the plate,` `pick up the black bowl on the cookie box and place it on the plate`
LIBERO-Object	`pick up the tomato sauce and place it in the basket,` `pick up the chocolate pudding and`

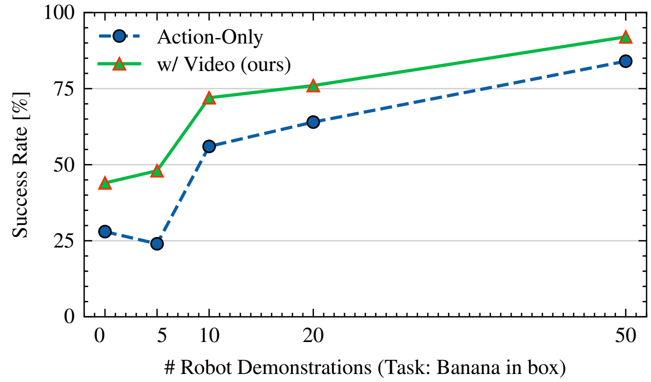

Fig. 8: Real world tasks scaling.

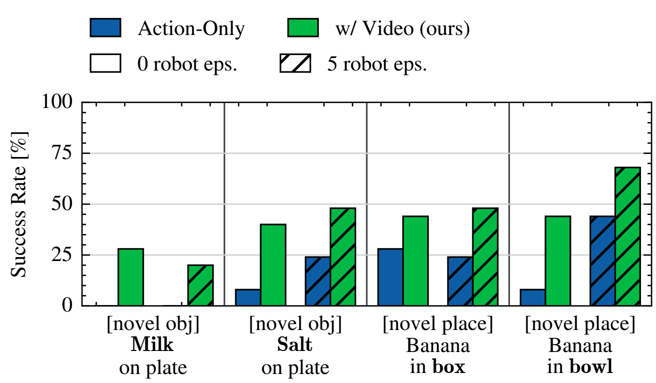

Fig. 9: Real world tasks all tasks.

Appendix D
Training and Model Parameters

TABLE IX: Model and training hyperparameters of the two stages.

Latent action model	
Encoder Φ	trained from scratch
Hidden dim d	1024
Spatial layers	8
Temporal layers	8
FDM decoder layers	8 (cross-attend z)
Bottleneck B	1024 $\rightarrow$ 8, strided conv
Input resolution	256×256
Latent shape	1×8
Params	344M
Frame window Δ	$\{0, 1, 5, 9\}$
Reconstruction loss ρ	Huber, β=0.1
Steps	70k
Optimizer	AdamW, bf16
Batch (per GPU $\times$ GPUs)	$32 \times 4 = 128$
Peak LR	1×10^{-4}
LR schedule	cosine, 1k warmup
Final LR	1×10^{-6}
Grad-norm clip	1.0
Policy	
Backbone	SmolVLM2-500M (frozen)
Backbone layers	16 of 32 (layer skipping)
Backbone hidden dim	960
Expert layers	16 (each expert)
Expert hidden dim	720
Input resolution	512×512
Params (trainable / total)	202M / 552M
Chunk K / plan M	50 / 10 (stride H=5)
Flow time τ	Beta(1.5, 1.0)
Euler steps	10
Loss weights	λ_a=λ_z=1
Teacher forcing p	1.0 $\rightarrow$ 0.2
Steps	100k
Optimizer	AdamW, bf16
Batch (per GPU $\times$ GPUs)	$80 \times 1 = 80$
Peak LR	1×10^{-4}
LR schedule	cosine, 100 warmup
Final LR	2.5×10^{-6}
Grad-norm clip	10.0

References

[1] K. Black et al., “$\pi_{0.5}$: A vision-language-action model with open-world generalization,” in *Proceedings of the 9th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, PMLR, 2025.

[2] K. Black et al., “π_0: A vision-language-action flow model for general robot control,” in *Proceedings of Robotics: Science and Systems (RSS)*, Los Angeles, CA, USA, 2025.

[3] D. Schmidt and M. Jiang, “Learning to act without actions,” in *International Conference on Learning Representations (ICLR)*, 2024.

[4] J. Bruce et al., “Genie: Generative interactive environments,” in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, PMLR, 2024.

[5] S. Ye et al., “Latent action pretraining from videos,” in *International Conference on Learning Representations (ICLR)*, 2025.

[6] X. Chen et al., “villa-X: Enhancing latent action modeling in vision-language-action models,” in *International Conference on Learning Representations (ICLR)*, 2026.

[7] A. Liang, P. Czempin, M. Hong, Y. Zhou, E. Biyik, and S. Tu, “CLAM: Continuous latent action models for robot learning from unlabeled demonstrations,” in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[8] J. Yang et al., “CoMo: Learning continuous latent motion from internet videos for scalable robot learning,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

[9] A. Nikulin et al., “Latent action learning requires supervision in the presence of distractors,” in *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025.

[10] X. Bu, J. Lyu, F. Sun, R. Yang, Z. Ma, and W. Li, “LAOF: Robust latent action learning with optical flow constraints,” *arXiv preprint arXiv:2511.16407*, 2026.

[11] Q. Bu et al., “UniVLA: Learning to act anywhere with task-centric latent actions,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2025.

[12] B. Liu et al., “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

[13] S. Fei et al., “LIBERO-Plus: In-depth robustness analysis of vision-language-action models,” *arXiv preprint arXiv:2510.13626*, 2025.

[14] A. Mandlekar et al., “MimicGen: A data generation system for scalable robot learning using human demonstrations,” in *Proceedings of the 7th Conference on Robot Learning (CoRL)*, 2023.

[15] Q. Li et al., “RotVLA: Rotational latent action for vision-language-action model,” *arXiv preprint arXiv:2605.13403*, 2026.

[16] J. A. Collins, L. Cheng, K. Aneja, A. Wilcox, B. Joffe, and A. Garg, “AMPLIFY: Actionless motion priors for robot learning from videos,” in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[17] M. J. Kim et al., “OpenVLA: An open-source vision-language-action model,” in *Proceedings of the 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, PMLR, 2025.

[18] Octo Model Team et al., “Octo: An open-source generalist robot policy,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2024.

[19] Y. Ze, G. Zhang, K. Zhang, C. Hu, M. Wang, and H. Xu, “3D diffusion policy: Generalizable visuomotor policy learning via simple 3D representations,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2024.

[20] A. Goyal, V. Blukis, J. Xu, Y. Guo, Y.-W. Chao, and D. Fox, “RVT-2: Learning precise manipulation from few demonstrations,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2024.

[21] S. Qian, K. Mo, V. Blukis, D. F. Fouhey, D. Fox, and A. Goyal, “3D-MVP: 3D multiview pretraining for manipulation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[22] F. Liu, F. Yan, L. Zheng, C. Feng, Y. Huang, and L. Ma, “RoboUniView: Visual-language model with unified view representation for robotic manipulation,” *arXiv preprint arXiv:2406.18977*, 2024.

[23] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, “PointNet: Deep learning on point sets for 3D classification and segmentation,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[24] I.-C. A. Liu, K. Choromanski, S. Huang, and C. Schenck, “CLAMP: Contrastive learning for 3D multi-view action-conditioned robotic manipulation pretraining,” *arXiv preprint arXiv:2602.00937*, 2026.

[25] S. Jeong et al., “Learning multi-view spatial reasoning from cross-view relations,” *arXiv preprint arXiv:2603.27967*, 2026.

[26] Y. Seo, J. Kim, S. James, K. Lee, J. Shin, and P. Abbeel, “Multi-view masked world models for visual robotic manipulation,” in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, PMLR, 2023.

[27] J.-C. Pang et al., “Learning view-invariant world models for visual robotic manipulation,” in *International Conference on Learning Representations (ICLR)*, 2025.

[28] J. M. Lee et al., “MVP-LAM: Learning action-centric latent action via cross-viewpoint reconstruction,” *arXiv preprint arXiv:2602.03668*, 2026.

[29] L. Wang, X. Chen, J. Zhao, and K. He, “Scaling proprioceptive-visual learning with heterogeneous pretrained transformers,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[30] R. Doshi, H. R. Walke, O. Mees, S. Dasari, and S. Levine, “Scaling cross-embodied learning: One policy for manipulation, navigation, locomotion and aviation,” in *Proceedings of the 8th Conference on Robot Learning*

(CoRL), ser. Proceedings of Machine Learning Research, PMLR, 2025.

[31] NVIDIA et al., "GR00T N1: An open foundation model for generalist humanoid robots," *arXiv preprint arXiv:2503.14734*, 2025.

[32] L. Y. Chen, K. Hari, K. Dharmarajan, C. Xu, Q. Vuong, and K. Goldberg, "Mirage: Cross-embodiment zero-shot policy transfer with cross-painting," in *Proceedings of Robotics: Science and Systems (RSS)*, Delft, Netherlands, 2024.

[33] L. Y. Chen et al., "RoVi-Aug: Robot and viewpoint augmentation for cross-embodiment robot learning," in *Proceedings of the 8th Conference on Robot Learning (CoRL)*, Munich, Germany, 2024.

[34] M. Lepert, R. Doshi, and J. Bohg, "SHADOW: Leveraging segmentation masks for cross-embodiment policy transfer," in *Proceedings of the 8th Conference on Robot Learning (CoRL)*, Munich, Germany, 2024.

[35] M. Lepert, J. Fang, and J. Bohg, "Phantom: Training robots without robots using only human videos," in *Proceedings of the 9th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, PMLR, 2025.

[36] S. Kareer et al., "EgoMimic: Scaling imitation learning via egocentric video," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[37] T. Tao, M. K. Srirama, J. J. Liu, K. Shaw, and D. Pathak, "DexWild: Dexterous human interactions for in-the-wild robot policies," in *Proceedings of Robotics: Science and Systems (RSS)*, 2025.

[38] K. Zakka, A. Zeng, P. Florence, J. Tompson, J. Bohg, and D. Dwibedi, "XIRL: Cross-embodiment inverse reinforcement learning," in *Proceedings of the 5th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, PMLR, 2022.

[39] M. Xu, Z. Xu, C. Chi, M. Veloso, and S. Song, "XSkill: Cross embodiment skill discovery," in *Proceedings of the 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, PMLR, 2023.

[40] J. Zheng et al., "Universal actions for enhanced embodied foundation models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[41] E. Bauer, E. Nava, and R. K. Katzschmann, "Latent action diffusion for cross-embodiment manipulation," *arXiv preprint arXiv:2506.14608*, 2025.

[42] H. Kim, J. Kang, H. Kang, M. Cho, S. J. Kim, and Y. Lee, "UniSkill: Imitating human videos via cross-embodiment skill representations," in *Proceedings of the 9th Conference on Robot Learning (CoRL)*, 2025.

[43] Open X-Embodiment Collaboration et al., "Open X-Embodiment: Robotic learning datasets and RT-X models," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

[44] X. Chen et al., "IGOR: Image-GOal representations are the atomic control units for foundation models in embodied AI," *arXiv preprint arXiv:2411.00785*, 2024.

[45] G. Bertasius, H. Wang, and L. Torresani, "Is space-time attention all you need for video understanding?" In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, PMLR, 2021.

[46] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lučić, and C. Schmid, "ViViT: A video vision transformer," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[47] Z. Liu et al., "Swin transformer V2: Scaling up capacity and resolution," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[48] J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, and Y. Liu, "RoFormer: Enhanced transformer with rotary position embedding," *Neurocomputing*, 2024.

[49] H. Su, S. Maji, E. Kalogerakis, and E. Learned-Miller, "Multi-view convolutional neural networks for 3D shape recognition," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015.

[50] C. Chi et al., "Diffusion policy: Visuomotor policy learning via action diffusion," in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

[51] M. Shukor et al., "SmolVLA: A vision-language-action model for affordable and efficient robotics," *arXiv preprint arXiv:2506.01844*, 2025.

[52] S. Bengio, O. Vinyals, N. Jaitly, and N. Shazeer, "Scheduled sampling for sequence prediction with recurrent neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.

[53] Y. Zhu et al., "robosuite: A modular simulation framework and benchmark for robot learning," *arXiv preprint arXiv:2009.12293*, 2020.

[54] The Robot Learning Company, *The robot learning company: TRLC-DK1*, https://www.robot-learning.co/, Accessed: 2026-07-28.